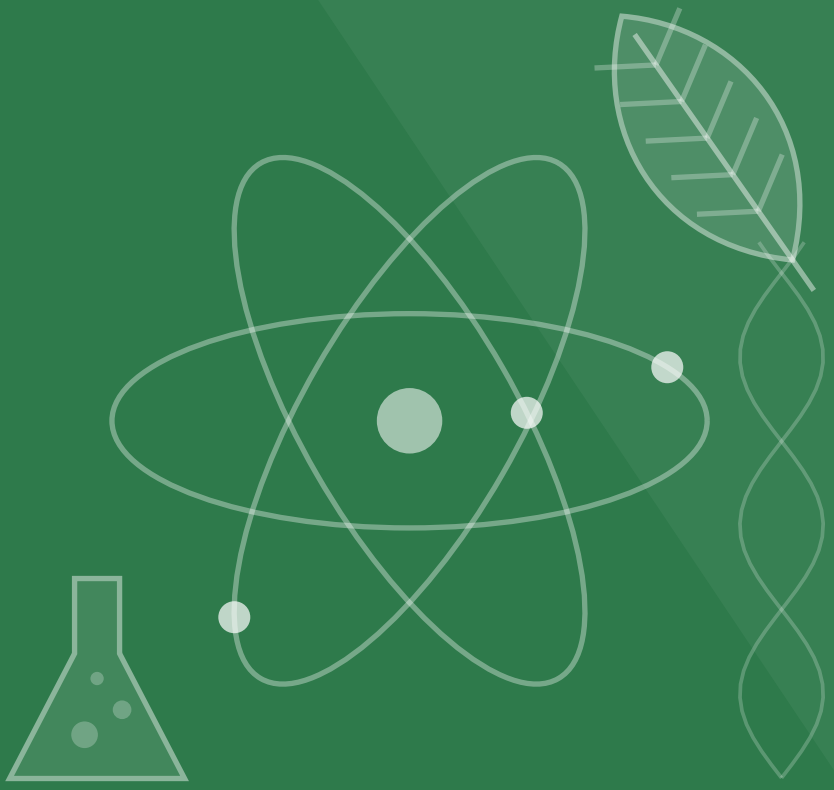
VENVLEARNING

STUDENT TEXTBOOK



Science

GRADE 4

Lessons, examples, and practice for the whole year

2 units

A VenvLearning family homeschool curriculum

GRADE 4 · SCIENCE

VENVLEARNING

Grade 4
Science

STUDENT TEXTBOOK

Unit 1 The Human Body

Unit 2 Energy, Heat, and Light

A VenvLearning family homeschool curriculum

How to use this book

Every lesson starts on a new page with the same parts, so you always know where to look. Read the Explain It boxes to see how a scientist reasons from evidence, and answer the Check Your Understanding questions. Your practice pages are in your Student Workbook, labeled with the same lesson number.

- Today's Goal (the box with the target) says what you will be able to do by the end of the lesson.
- Today's Assignment is your checklist for the day. Tick each box as you finish it.
- Each lesson page has three panels. The wide center panel is the lesson, in numbered steps: 1 Get Ready, 2 Learn, and 3 Check Your Understanding.
- The left panel holds things to remember: Key Ideas (the most important rule or fact; learn these), Watch Out warnings about common mistakes, tips, and a Remember note that sums up the lesson. Each one sits beside the part of the lesson it goes with.
- The right panel holds Words to Know (definitions), what to read in your Reader, and other extras.
- Pages that look like a page from a book, on cream paper with a brown READING tab, are things to read: stories, excerpts, and articles.
- Investigate boxes are hands-on experiments. Do them with your teacher and record what you observe.
- At the end of each week of lessons there is a short Weekly Check in your workbook. Each unit ends with the Unit Test.

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UNIT 1

The Human Body

Thinking like a scientist; cells, tissues, and organs; and the body systems that keep you alive.

Your body is an amazing machine made of trillions of cells. In this unit you'll learn to think like a scientist: observing, asking testable questions, running fair tests, and using evidence. Then you'll use those skills to explore how your body is organized and how its systems work together.

In this unit you will…

- Tell observations from inferences, and plan fair tests.
- Support claims with evidence and reasoning (CER).
- Describe levels of organization from cells to systems.
- Explain the jobs of the digestive, respiratory, circulatory, skeletal, muscular, and nervous systems and how they work together.

SCIENCE IN YOUR WORLD

Doctors, nurses, athletes, and coaches use what they know about body systems every day to keep people healthy.

UNIT VOCABULARY

observation	Information gathered with senses or tools.
inference	A conclusion based on observations.
fair test	Only one variable is changed.
cell	The smallest living unit.
organ system	Organs that work together.
nutrients	Substances in food the body uses.

Lessons in this unit

Lesson	Title
1.1	Thinking Like a Scientist
1.2	Fair Tests and Evidence
1.3	Cells, Tissues, Organs, and Systems
1.4	Body Systems Work Together
1.5	The Digestive System
1.6	The Respiratory System
1.7	The Heart and Blood Vessels
1.8	Bones, Muscles, and Healthy Choices

REMEMBER

LESSON 1.1
Week 1 · Lesson 1 of 2
Unit 1: The Human Body

Lesson 1.1: Thinking Like a Scientist

TODAY'S GOAL

I can tell the difference between an observation and an inference, and ask a question that can be investigated.

Standards: Science and Engineering Practices 1, 3

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.1 in this book with your teacher.
- [ ] Do the Investigate activity: Mystery Bag.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.1, Part A.
- [ ] Complete the rest of Workbook Lesson 1.1 (Parts B–D).

1 Get Ready

I WONDER

You walk into the kitchen and see crumbs on the counter and an empty cookie jar. What do you SEE? What do you THINK happened?

2 Learn

Scientists are curious people who ask questions about the world and then look for evidence to answer them. Two of their most important tools are observations and inferences.

WORDS & MORE

WORDS TO KNOW

observation
Information you gather with your senses or with tools.

inference
A conclusion you reach from observations and what you already know.

investigate
To find out about something by observing, testing, and measuring.

REMEMBER

Observation	Inference
what you notice with your senses or tools	what you figure out from what you noticed
"The cookie jar is empty."	"Someone ate the cookies."
"The ground is wet."	"It rained last night."
"The thermometer reads 38°F."	"It is too cold to swim outside."

An inference can be wrong. Maybe the cookies were moved to another jar! Good scientists gather more observations to test their inferences.

WORDS & MORE

EXPLAIN IT · ASK A TESTABLE QUESTION

You notice that a puddle in the sun dries faster than a puddle in the shade.

1 Observation: the sunny puddle disappeared first.

2 Question that can't be tested easily: "Why does water exist?"

3 Testable question: "Does water evaporate faster in sunlight than in shade?" You can set up two cups of water and measure.

Explanation: A testable question can be answered by observing, measuring, or experimenting.

KEY IDEA

Observations come from your senses and tools. Inferences are explanations based on observations. Testable questions can be answered with evidence.

INVESTIGATE · MYSTERY BAG

What can you figure out about an object you can't see?

You need: a paper bag with a hidden object (chosen by your teacher); paper and a pencil.

1 Without looking, shake the bag and listen. Feel the object through the paper.

2 List at least four observations (sound, weight, shape, texture).

3 Write one inference about what the object is, and the observations that support it.

4 Open the bag. Was your inference correct?

Record: a two-column chart: Observations | Inferences.

✓ REMEMBER

Observations come from senses and tools; inferences explain them; testable questions can be answered with evidence.

REMEMBER

WORDS & MORE

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Write observation or inference for each sentence, or do what the item asks. Write your answers in Workbook Lesson 1.1, Part A. Then check with your teacher.

1. Observation or inference: “The leaf is green and has jagged edges.”
2. Observation or inference: “The dog is hungry because it is barking at its bowl.”
3. Make this question testable: “Why do plants like light?”

REMEMBER

LESSON 1.2
Week 1 · Lesson 2 of 2
Unit 1: The Human Body

Lesson 1.2: Fair Tests and Evidence

TODAY'S GOAL

I can plan a fair test and use evidence to support a claim.

Standards: Science and Engineering Practices 3, 4, 7

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.2 in this book with your teacher.
- [ ] Do the Investigate activity: The Paper Towel Test.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.2, Part A.
- [ ] Complete the rest of Workbook Lesson 1.2 (Parts B–D).

1 Get Ready

I WONDER

Two friends race paper airplanes. One throws hard and one throws gently. Is it a fair race? Why or why not?

2 Learn

In a fair test, you change only one thing, the variable you test, and keep everything else the same. Then you measure what happens. If you change two things at once, you can't tell which one caused the result.

WORDS & MORE

WORDS TO KNOW

variable
Something that can change in an investigation.

fair test
An investigation where only one variable is changed on purpose.

evidence
Observations and data that support a claim.

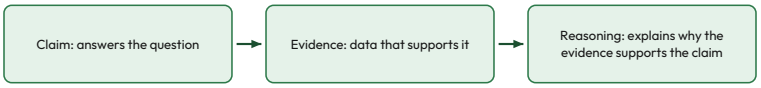
claim
A statement that answers a question.

REMEMBER

WORDS & MORE

Question	Change (test)	Keep the same	Measure
Which paper towel brand absorbs the most water?	brand of towel	towel size, amount of water, time	water left in the cup (mL)

After a test, scientists write a claim, support it with evidence (data), and give reasoning that explains why the evidence supports the claim. This is called CER.



KEY IDEA

Change one variable, keep the rest the same, measure the result, and support your claim with evidence and reasoning.

INVESTIGATE · THE PAPER TOWEL TEST

Which paper towel absorbs the most water?

You need: 2 or 3 kinds of paper towel cut to the same size; measuring cup; water; 3 cups.

1. Pour 100 mL of water into each cup.
2. Dip one towel square in each cup for 10 seconds.
3. Lift the towel and let it drip back into the cup for 10 seconds.
4. Measure the water left in each cup.

Record: a data table, then a CER: Which towel absorbed the most? What is your evidence?

✓ REMEMBER

A fair test changes one variable; CER = claim, evidence, reasoning.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.2, Part A. Then check with your teacher.

1. Why must the towel squares be the same size?
2. What is the claim in CER?
3. A student tests plant growth but gives one plant more water AND more light. What is wrong?

REMEMBER

LESSON 1.3
Week 2 · Lesson 1 of 2
Unit 1: The Human Body

Lesson 1.3: Cells, Tissues, Organs, and Systems

TODAY'S GOAL

I can describe how the body is organized from cells to organ systems.

Standards: 4-LS1-1

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.3 in this book with your teacher.
- [ ] Do the Investigate activity: Build the Levels.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.3, Part A.
- [ ] Complete the rest of Workbook Lesson 1.3 (Parts B–D).

1 Get Ready

I WONDER

A brick is part of a wall, a wall is part of a room, and a room is part of a house. How might your body be built the same way?

2 Learn

Your body is made of trillions of tiny cells, so small you need a microscope to see them. Similar cells work together as tissues, such as muscle tissue. Different tissues form organs, like the heart or stomach. Organs work together in organ

WORDS & MORE

WORDS TO KNOW

cell
The smallest living unit of a living thing.

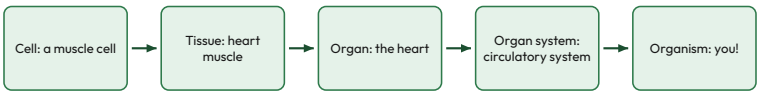
tissue
A group of similar cells that work together.

organ
A body part made of tissues that does a special job.

organ system
A group of organs that work together.

REMEMBER

systems.



There are many kinds of cells. Nerve cells are long and thin to carry messages. Red blood cells are round and flat to carry oxygen. A cell's shape fits its job.

KEY IDEA

Cells → tissues → organs → organ systems → organism.

INVESTIGATE · BUILD THE LEVELS

How can a model show levels of organization?

You need: building blocks or LEGO bricks; sticky notes.

1. Let one brick stand for one cell.
2. Join several matching bricks into a “tissue.”
3. Combine different “tissues” into an “organ.”
4. Put several organs together into a “system.” Label each level with a sticky note.

Record: a sketch of your model with each level labeled.

✓ REMEMBER

Cells form tissues, tissues form organs, organs form systems.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.3, Part A. Then check with your teacher.

1. What is the smallest living unit of the body?
2. Put in order: organ, cell, organ system, tissue
3. Name one organ.

WORDS & MORE

REMEMBER

LESSON 1.4
Week 2 · Lesson 2 of 2
Unit 1: The Human Body

Lesson 1.4: Body Systems Work Together

TODAY'S GOAL

I can name major body systems, describe each one's main job, and explain how they work together.

Standards: 4-LS1-1

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.4 in this book with your teacher.
- [ ] Do the Investigate activity: Systems Relay.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.4, Part A.
- [ ] Complete the rest of Workbook Lesson 1.4 (Parts B–E).

1 Get Ready

I WONDER

Think about kicking a soccer ball. Which parts of your body are working?

2 Learn

WORDS & MORE

WORDS TO KNOW

skeletal system
Bones that support and protect the body.

muscular system
Muscles that move the body.

nervous system
The brain, spinal cord, and nerves that control the body.

digestive system
Organs that break down food.

respiratory system
Organs that bring in oxygen and remove carbon dioxide.

circulatory system
The heart and blood vessels that move blood.

REMEMBER

System	Main organs	Main job
skeletal	bones	supports, protects, works with muscles to move
muscular	muscles	moves bones and organs
nervous	brain, spinal cord, nerves	senses, thinks, sends messages
digestive	mouth, stomach, intestines	breaks food into nutrients
respiratory	nose, windpipe, lungs	takes in oxygen, gets rid of carbon dioxide
circulatory	heart, blood vessels, blood	carries oxygen and nutrients to cells

WORDS & MORE

EXPLAIN IT · SYSTEMS WORKING TOGETHER

Kicking a soccer ball

1. Nervous: your eyes see the ball and your brain sends a message to your leg.
2. Muscular and skeletal: leg muscles pull on bones to swing your leg.
3. Respiratory and circulatory: you breathe faster and your heart pumps faster to deliver oxygen to your muscles.
4. Digestive: nutrients from food give your cells energy.

Explanation: No system works alone; all of them work together to keep you alive and moving.

KEY IDEA

Each body system has a main job, and the systems depend on one another.

INVESTIGATE · SYSTEMS RELAY

Which systems work when you do an everyday task?

You need: index cards labeled with the six systems; a timer.

1. Choose an action: eating an apple, running up stairs, or reading a book aloud.
2. Do the action (safely!).
3. Hold up every system card you think was working, and explain why.

Record: the action and the systems used, with one reason for each.

✓ REMEMBER

Six major systems each have a job, and they work together.

REMEMBER

WORDS & MORE

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.4, Part A. Then check with your teacher.

1. Which system carries oxygen to cells?
2. Which system breaks down food?
3. Which systems help you jump?

REMEMBER

LESSON 1.5
Week 3 · Lesson 1 of 2
Unit 1: The Human Body

Lesson 1.5: The Digestive System

TODAY'S GOAL

I can describe the path of food through the digestive system and explain what each organ does.

Standards: 4-LS1-1

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.5 in this book with your teacher.
- [ ] Do the Investigate activity: Stomach in a Bag.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.5, Part A.
- [ ] Complete the rest of Workbook Lesson 1.5 (Parts B–D).

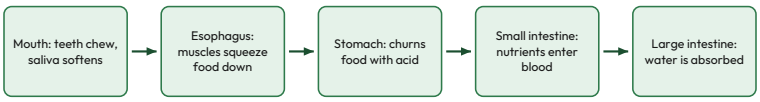
1 Get Ready

I WONDER

Where does your breakfast go after you swallow it?

2 Learn

Your body can't use a sandwich as a sandwich. The digestive system breaks food into tiny nutrients that can pass into your blood and travel to your cells.



WORDS & MORE

WORDS TO KNOW

nutrients
Substances in food that the body uses for energy, growth, and repair.

esophagus
The tube that carries food from the mouth to the stomach.

small intestine
The long, coiled tube where most nutrients pass into the blood.

large intestine
The wider tube where water is absorbed from leftover food.

REMEMBER

WORDS & MORE

Digestion is both mechanical (chewing and churning break food into smaller pieces) and chemical (saliva, stomach acid, and other juices break food down into nutrients). The liver and pancreas make juices that help in the small intestine. In an adult, the small intestine is about 20 feet long, coiled up to fit inside the body!

KEY IDEA

Food travels mouth → esophagus → stomach → small intestine → large intestine. Nutrients enter the blood in the small intestine.

INVESTIGATE · STOMACH IN A BAG

How does the stomach help break down food?

You need: zip-top plastic bag; 4 crackers; 1/2 cup water; a few drops of vinegar (to stand for acid).

1. Put the crackers in the bag. Observe them.
2. Add the water and vinegar and seal the bag tightly.
3. Gently squeeze and mash the bag for 2 minutes, the way stomach muscles churn.
4. Observe the crackers again.

Record: before-and-after observations, and which part of digestion the squeezing models.

Safety: Do not eat or drink anything used in an investigation.

✓ REMEMBER

Digestion breaks food into nutrients that enter the blood in the small intestine.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.5, Part A. Then check with your teacher.

1. Where do most nutrients enter the blood?
2. What does the large intestine absorb?
3. What did squeezing the bag model?

REMEMBER

LESSON 1.6
Week 3 · Lesson 2 of 2
Unit 1: The Human Body

Lesson 1.6: The Respiratory System

TODAY'S GOAL

I can explain how the respiratory system brings oxygen into the body and removes carbon dioxide.

Standards: 4-LS1-1

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.6 in this book with your teacher.
- [ ] Do the Investigate activity: Breathing Rate.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.6, Part A.
- [ ] Complete the rest of Workbook Lesson 1.6 (Parts B–E).

1 Get Ready

I WONDER

Hold your breath for a few seconds. Why can't you hold it forever?

2 Learn

Every cell in your body needs oxygen to release energy from nutrients. Cells also make carbon dioxide, a waste gas. The respiratory system brings oxygen in and sends carbon dioxide out.

WORDS & MORE

WORDS TO KNOW

oxygen
A gas in air that cells need to get energy from food.

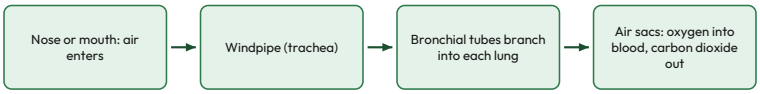
carbon dioxide
A waste gas your cells make and you breathe out.

diaphragm
A dome-shaped muscle below the lungs that helps you breathe.

air sacs
Tiny balloon-like sacs in the lungs where gases pass into and out of the blood.

REMEMBER

WORDS & MORE



When the diaphragm tightens and moves down, the chest gets bigger and air rushes into the lungs. When it relaxes and moves up, air is pushed out. Inside the lungs are millions of tiny air sacs wrapped in blood vessels, where oxygen moves into the blood and carbon dioxide moves out.

KEY IDEA

You breathe in oxygen that cells need and breathe out carbon dioxide that cells make. Gases are exchanged in the air sacs.

INVESTIGATE · BREATHING RATE

How does exercise change how fast you breathe?

You need: a timer; a partner.

1. Sit quietly for 1 minute. Count your breaths for 30 seconds and multiply by 2.
2. Do jumping jacks for 1 minute.
3. Right away, count your breaths for 30 seconds and multiply by 2.
4. Rest 3 minutes and count again.

Record: breaths per minute at rest, after exercise, and after resting, in a table and a bar graph.

✓ REMEMBER

The respiratory system takes in oxygen and removes carbon dioxide.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.6, Part A. Then check with your teacher.

1. What gas do your cells need?
2. What muscle helps you breathe?
3. Why does breathing speed up during exercise?

REMEMBER

LESSON 1.7
Week 4 · Lesson 1 of 2
Unit 1: The Human Body

Lesson 1.7: The Heart and Blood Vessels

TODAY'S GOAL

I can describe how the heart and blood vessels move blood through the body, and measure my pulse.

Standards: 4-LS1-1

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.7 in this book with your teacher.
- [ ] Do the Investigate activity: Take Your Pulse.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.7, Part A.
- [ ] Complete the rest of Workbook Lesson 1.7 (Parts B–D).

1 Get Ready

I WONDER

Make a fist. Your heart is about that size. How many times do you think it beats in one minute?

2 Learn

The heart is a muscle about the size of your fist. It has four chambers and pumps blood through a huge network of blood vessels. Arteries carry blood away from the heart. Capillaries are tiny vessels where oxygen and nutrients move into cells and wastes move out. Veins carry blood back to the heart.

WORDS & MORE

WORDS TO KNOW

heart
A muscular organ that pumps blood.

artery
A blood vessel that carries blood away from the heart.

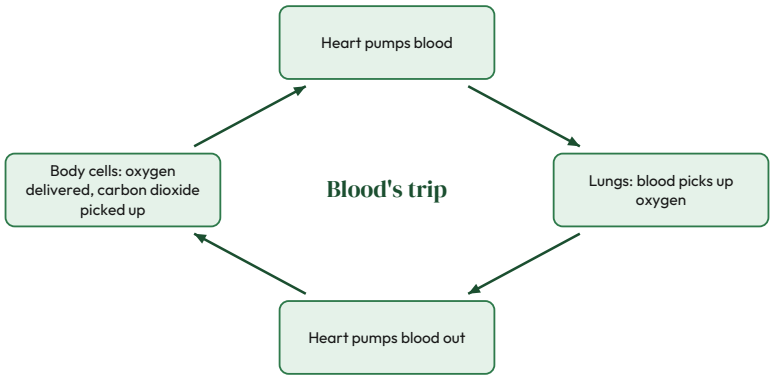
vein
A blood vessel that carries blood back to the heart.

capillary
A tiny blood vessel where materials pass between blood and cells.

pulse
The beat you can feel as the heart pushes blood through an artery.

REMEMBER

WORDS & MORE



KEY IDEA

The heart pumps blood through arteries to capillaries and back through veins, delivering oxygen and nutrients and carrying away wastes.

WATCH OUT

Blood in veins is not blue. Blood is always red; veins can look bluish through skin because of how light passes through it.

✓ REMEMBER

The heart pumps blood through arteries, capillaries, and veins.

INVESTIGATE · TAKE YOUR PULSE

How does activity change heart rate?

You need: a timer.

1. Place two fingers (not your thumb) on the side of your neck or the inside of your wrist.
2. Count the beats for 15 seconds. Multiply by 4 to find beats per minute.
3. Run in place for 1 minute, then measure again.
4. Rest 3 minutes and measure again.

Record: beats per minute in a table; compare with your breathing data from Lesson 1.6.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.7, Part A. Then check with your teacher.

1. Which vessels carry blood away from the heart?
2. Where do oxygen and nutrients move into cells?
3. 15-second count is 22 beats. Beats per minute?

REMEMBER

LESSON 1.8
Week 4 · Lesson 2 of 2
Unit 1: The Human Body

Lesson 1.8: Bones, Muscles, and Healthy Choices

TODAY'S GOAL

I can explain how bones and muscles work together and describe choices that keep body systems healthy.

Standards: 4-LS1-1

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.8 in this book with your teacher.
- [ ] Do the Investigate activity: Muscle Partners.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.8, Part A.
- [ ] Complete the rest of Workbook Lesson 1.8 (Parts B–E).

1 Get Ready

I WONDER

Bend and straighten your arm. What do you feel happening in your upper arm?

2 Learn

An adult skeleton has 206 bones. Bones support the body, protect organs (the skull protects the brain; the ribs protect the heart and lungs), and make blood cells inside. Muscles are attached to bones by tendons. Muscles can only pull, not push, so they work in pairs.

WORDS & MORE

WORDS TO KNOW

joint
A place where two bones meet.

tendon
A tough cord that connects a muscle to a bone.

ligament
A tough band that connects bone to bone.

REMEMBER

WORDS & MORE

EXPLAIN IT · MUSCLE PAIRS

Bending and straightening your arm

1 To bend: the biceps (front of upper arm) tightens and pulls the forearm up; the triceps relaxes.

2 To straighten: the triceps (back of upper arm) tightens and pulls; the biceps relaxes.

Explanation: Muscles work in pairs because each muscle can only pull.

Healthy choice	Systems it helps
active play about 60 minutes a day	muscular, skeletal, circulatory, respiratory
foods like fruits, vegetables, whole grains, and protein	digestive (and every system that needs nutrients)
9–12 hours of sleep a night for children your age	nervous (and all systems repair during sleep)
drinking water	digestive, circulatory
wearing a helmet and seat belt	skeletal, nervous

KEY IDEA

Bones support and protect; muscles pull in pairs to move bones at joints. Healthy habits keep all systems working.

INVESTIGATE · MUSCLE PARTNERS

How do muscles work in pairs?

You need: your own arm.

1 Place your hand on your biceps. Slowly bend your arm. What does the muscle do?

2 Move your hand to your triceps. Straighten your arm. What does that muscle do?

Record: a labeled sketch of the upper arm showing which muscle pulls to bend and which pulls to straighten.

✓ REMEMBER

Bones support and protect; muscles pull in pairs; healthy habits keep systems strong.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.8, Part A. Then check with your teacher.

1. How many bones in an adult skeleton?
2. What connects muscle to bone?
3. Why do muscles work in pairs?

UNIT 1 REVIEW

Unit 1 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
1.1	Observations come from senses and tools; inferences explain them; testable questions can be answered with evidence.
1.2	A fair test changes one variable; CER = claim, evidence, reasoning.
1.3	Cells form tissues, tissues form organs, organs form systems.
1.4	Six major systems each have a job, and they work together.
1.5	Digestion breaks food into nutrients that enter the blood in the small intestine.
1.6	The respiratory system takes in oxygen and removes carbon dioxide.
1.7	The heart pumps blood through arteries, capillaries, and veins.
1.8	Bones support and protect; muscles pull in pairs; healthy habits keep systems strong.

UNIT 2

Energy, Heat, and Light

Forms of energy, heat and temperature, conduction, convection, and radiation, conductors and insulators, how light travels, reflection, refraction, and color.

Energy is everywhere: in a rolling ball, a warm mug, a sunny window, and a rainbow. In this unit you'll discover what energy is, measure temperature, watch heat move in three different ways, and design a way to keep ice from melting. Then you'll explore light: how it travels, how we see, why shadows form, and how a glass of water can make a rainbow.

In this unit you will...

- Describe forms of energy and relate speed to energy.
- Measure temperature and explain how heat moves.
- Compare conductors and insulators and design a solution.
- Model how light travels, reflects, refracts, and lets us see.

SCIENCE IN YOUR WORLD

Engineers design insulation for homes and coolers, eye doctors use lenses to help people see, and solar engineers capture the sun's light and heat for energy.

UNIT VOCABULARY

energy	The ability to make something move or change.
heat	Energy flowing from warmer to cooler.
conductor	Lets heat move easily.
insulator	Slows heat.
reflect	Bounce off.
refraction	The bending of light.

Lessons in this unit

Lesson	Title
2.1	What Is Energy?
2.2	Heat and Temperature
2.3	How Heat Moves
2.4	Conductors and Insulators
2.5	Light Travels and How We See
2.6	Reflection, Absorption, and Transparency
2.7	Refraction and Color
2.8	Light Investigation: Claim, Evidence, and Reasoning

REMEMBER

LESSON 2.1
Week 1 · Lesson 1 of 2
Unit 2: Energy, Heat, and Light

Lesson 2.1: What Is Energy?

TODAY'S GOAL

I can describe forms of energy and explain that faster objects have more energy.

Standards: 4-PS3-1, 4-PS3-2

TODAY'S ASSIGNMENT

- [ ] Read Lesson 2.1 in this book with your teacher.
- [ ] Do the Investigate activity: Ramp and Cup.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.1, Part A.
- [ ] Complete the rest of Workbook Lesson 2.1 (Parts B–E).

1 Get Ready

I WONDER

A bowling ball rolls slowly into the pins. Then it rolls fast. Which time do more pins fall? Why?

2 Learn

Energy is the ability to make something move or change. You can't hold energy in your hand, but you can see what it does: a ball rolls, a lamp glows, a pot of soup gets hot, a drum makes sound.

WORDS & MORE

WORDS TO KNOW

energy
The ability to make something move or change.

motion energy
The energy an object has because it is moving.

transfer
To move from one place or object to another.

energy source
Something that gives off energy, such as the sun or a battery.

REMEMBER

WORDS & MORE

Form of energy	What it does	Example
motion	makes things move	a rolling ball, wind, a running dog
sound	travels as vibrations	a drum, a voice, thunder
light	lets us see	the sun, a flashlight, a candle
heat (thermal)	makes things warmer	a stove, a campfire, your hands rubbing together
electrical	flows through wires	a battery, an outlet, lightning

Energy can transfer, or move, from one object to another. When a moving ball hits another ball, some of its motion energy moves into the second ball. Energy can also change form: a flashlight changes electrical energy into light and a little heat.

The faster an object moves, the more motion energy it has. A fast ball hits harder, makes a louder sound, and pushes other objects farther than a slow ball of the same size.

EXPLAIN IT · USE EVIDENCE ABOUT SPEED AND ENERGY

A marble rolled down a steep ramp knocks a cup 30 cm. The same marble rolled down a low ramp knocks it 10 cm.

1. The steep ramp made the marble move faster.
2. The faster marble pushed the cup farther.
3. A bigger push means more energy was transferred.

Explanation: The faster marble had more motion energy.

KEY IDEA

Energy makes things move or change. It comes in forms such as motion, sound, light, heat, and electrical, and it can transfer between objects. Faster objects have more motion energy.

INVESTIGATE · RAMP AND CUP

Does a faster marble have more energy?

You need: a ruler with a groove (or a cardboard ramp); a stack of books; a marble; a paper cup with a doorway cut out; a measuring tape.

1. Set the ramp on one book and place the cup at the bottom.
2. Let the marble go (don't push it) and measure how far the cup moves.
3. Repeat with two books, then three books.
4. Do three trials at each height and record the distances.

REMEMBER

✓ REMEMBER

Energy makes things move or change; faster objects have more motion energy.

Record: a data table of ramp height and distance, and a sentence that uses your data to answer the question.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 2.1, Part A. Then check with your teacher.

1. Name three forms of energy.
2. Which has more motion energy: a slow bike or a fast bike?
3. What energy changes happen in a flashlight?

WORDS & MORE

REMEMBER

LESSON 2.2
Week 1 · Lesson 2 of 2
Unit 2: Energy, Heat, and Light

Lesson 2.2: Heat and Temperature

TODAY'S GOAL

I can measure temperature and explain that heat flows from warmer objects to cooler ones.

Standards: 4-PS3-2

TODAY'S ASSIGNMENT

- [ ] Read Lesson 2.2 in this book with your teacher.
- [ ] Do the Investigate activity: Cooling Cups.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.2, Part A.
- [ ] Complete the rest of Workbook Lesson 2.2 (Parts B–E).

1 Get Ready

I WONDER

Why does a cup of hot cocoa get cold if you forget about it, but a glass of ice water gets warm?

2 Learn

Temperature tells how hot or cold something is. We measure it with a thermometer in degrees. In the United States we often use degrees Fahrenheit (°F); scientists use degrees Celsius (°C). Water freezes at 32°F (0°C) and boils at 212°F (100°C).

WORDS & MORE

WORDS TO KNOW

temperature
How hot or cold something is.

thermometer
A tool that measures temperature.

heat
Energy that flows from a warmer object to a cooler one.

degree
A unit for measuring temperature, shown with the symbol °.

REMEMBER

Heat is energy that moves from a warmer object to a cooler one, always in that direction. Your hot cocoa loses heat to the cooler air around it until they are the same temperature. Ice water gains heat from the warmer air. When you hold an ice cube, heat flows from your warm hand into the ice, which is why your hand feels cold.

WORDS & MORE

Temperature	°F	°C
water freezes	32°F	0°C
a cool fall day	50°F	10°C
room temperature	about 70°F	about 21°C
normal body temperature	about 98.6°F	about 37°C
water boils	212°F	100°C

EXPLAIN IT · READ A THERMOMETER

The red liquid in a thermometer stops at the line halfway between 60 and 70 on the °F side.

1. Find the two labeled numbers on each side of the top of the liquid: 60 and 70.
2. The liquid is halfway between them.
3. Halfway between 60 and 70 is 65.

Explanation: The temperature is 65°F.

KEY IDEA

Temperature is how hot or cold something is, measured with a thermometer. Heat is energy that always flows from warmer objects to cooler ones.

WATCH OUT

Cold is not something that flows into your hand. Heat flows out of your hand into the cold object. Heat always moves from warmer to cooler.

INVESTIGATE · COOLING CUPS

How does the temperature of warm water change over time?

You need: two cups; warm tap water and cold tap water; a thermometer; a timer.

1. Fill one cup with warm tap water and one with cold tap water.
2. Measure and record each temperature.
3. Measure both cups again every 5 minutes for 20 minutes.
4. Measure the room's air temperature too.

Record: a data table and a sentence explaining which way heat flowed for each cup.

Safety: Use warm tap water, not hot or boiling water.

REMEMBER

✓ REMEMBER

Temperature tells how hot something is; heat flows from warmer to cooler.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 2.2, Part A. Then check with your teacher.

1. What tool measures temperature?
2. At what temperature does water freeze in °F?
3. You hold a snowball. Which way does heat flow?

WORDS & MORE

REMEMBER

LESSON 2.3
Week 2 · Lesson 1 of 2
Unit 2: Energy, Heat, and Light

Lesson 2.3: How Heat Moves

TODAY'S GOAL

I can describe three ways heat moves: conduction, convection, and radiation.

Standards: 4-PS3-2

TODAY'S ASSIGNMENT

- [ ] Read Lesson 2.3 in this book with your teacher.
- [ ] Do the Investigate activity: Food Coloring Currents.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.3, Part A.
- [ ] Complete the rest of Workbook Lesson 2.3 (Parts B–D).

1 Get Ready

I WONDER

You sit near a campfire. Your face feels hot even though you aren't touching the fire. How does the heat reach you?

2 Learn

Heat moves in three ways. In conduction, heat moves through objects that are touching. A metal spoon in hot soup gets hot all the way up the handle as heat passes from particle to particle.

WORDS & MORE

WORDS TO KNOW

conduction
Heat moving through touching objects.

convection
Heat moving by the flow of a liquid or gas, such as air or water.

radiation
Heat moving through space as waves, such as heat from the sun or a fire.

REMEMBER

In convection, heat moves when a liquid or gas flows. Warm air and warm water rise, and cooler air and water sink, making a loop called a convection current. That's why the upstairs of a house is often warmer.

In radiation, heat travels as invisible waves, even through empty space. The sun's heat travels about 93 million miles through space to warm Earth. You feel radiation from a campfire or a toaster without touching it.

Way heat moves	How	Example
conduction	through touching objects	a spoon in hot soup; a pan on a stove burner
convection	by flowing liquid or gas	warm air rising from a heater; boiling water churning
radiation	as waves through space or air	sunshine on your skin; heat from a campfire

WORDS & MORE

KEY IDEA

Heat moves by conduction (touching), convection (flowing liquids and gases), and radiation (waves through space).

INVESTIGATE · FOOD COLORING CURRENTS

Can you see convection in water?

You need: a clear glass dish or jar of room-temperature water; a small cup of warm water with red food coloring; an ice cube made with blue food coloring.

1. Let the water in the dish sit until it is still.
2. Gently set the blue ice cube at one end.
3. Carefully pour a little red warm water in at the other end, near the bottom.
4. Watch from the side for 5 minutes. Where does each color go?

Record: a side-view drawing with arrows showing where the warm and cold water moved.

Safety: Use warm tap water only.

✓ REMEMBER

Heat moves by conduction, convection, and radiation.

REMEMBER

WORDS & MORE

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 2.3, Part A. Then check with your teacher.

1. How does the sun's heat reach Earth?
2. A pan gets hot on a burner. Which way does the heat move?
3. Why is the top floor of a house often warmer?

REMEMBER

LESSON 2.4
Week 2 · Lesson 2 of 2
Unit 2: Energy, Heat, and Light

Lesson 2.4: Conductors and Insulators

TODAY'S GOAL

I can test materials and design a way to slow the flow of heat.

Standards: 4-PS3-2, 4-PS3-4, 3-5-ETS1-1

TODAY'S ASSIGNMENT

- [ ] Read Lesson 2.4 in this book with your teacher.
- [ ] Do the Investigate activity: Ice Cube Keeper.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.4, Part A.
- [ ] Complete the rest of Workbook Lesson 2.4 (Parts B–E).

1 Get Ready

I WONDER

Why do pot handles often have plastic or wooden grips?

2 Learn

Some materials let heat move through them easily. They are conductors. Most metals, such as copper, aluminum, and iron, are good conductors. That's why pots and pans are made of metal: heat moves quickly from the burner to the food.

WORDS & MORE

WORDS TO KNOW

conductor
A material that lets heat move through it easily.

insulator
A material that slows the movement of heat.

design
A plan for building something that solves a problem.

criteria
What a design must do to succeed.

REMEMBER

WORDS & MORE

Other materials slow heat down. They are insulators. Wood, plastic, cloth, foam, and air trapped in tiny spaces are good insulators. A plastic pot handle keeps your hand from getting burned. A winter coat traps warm air near your body. A cooler keeps ice cold by slowing heat from coming in.

Good conductors	Good insulators
copper, aluminum, iron, steel	wood, plastic, rubber, foam
used to move heat quickly: pots, pans, radiators	used to slow heat: pot handles, coats, coolers, oven mitts

EXPLAIN IT · USE THE ENGINEERING DESIGN PROCESS

Problem: keep an ice cube from melting for as long as possible.

1. Ask: What are the criteria? The ice must stay frozen as long as possible, and the holder must fit in a cup.
2. Plan: Choose insulators such as foil-covered foam, cloth, or cotton balls.
3. Build and test: Time how long the ice lasts compared with an ice cube with no holder.
4. Improve: Change one thing and test again.

Explanation: Engineers ask, plan, build, test, and improve.

KEY IDEA

Conductors let heat move easily; insulators slow heat. Engineers choose materials to control the flow of heat.

INVESTIGATE · ICE CUBE KEEPER

Which design keeps an ice cube frozen longest?

You need: three ice cubes of the same size; three small cups; insulating materials: cloth, cotton balls, foam, newspaper, foil; a timer.

1. Leave one ice cube in a plain cup (your control).
2. Design and build two holders using different insulators.
3. Start the timer and check every 10 minutes.
4. Record when each ice cube is completely melted.

Record: a sketch of each design, your data table, and a sentence naming the best insulator.

✓ REMEMBER

Conductors move heat easily; insulators slow it; engineers design with both.

REMEMBER

WORDS & MORE

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 2.4, Part A. Then check with your teacher.

1. Is copper a conductor or an insulator?
2. Why are oven mitts made of thick cloth?
3. What are the steps of the design process?

REMEMBER

LESSON 2.5
Week 3 · Lesson 1 of 2
Unit 2: Energy, Heat, and Light

Lesson 2.5: Light Travels and How We See

TODAY'S GOAL

I can explain that light travels in straight lines and that we see objects when light reflects off them into our eyes.

Standards: 4-PS4-2

TODAY'S ASSIGNMENT

- [ ] Read Lesson 2.5 in this book with your teacher.
- [ ] Do the Investigate activity: Shadow Lengths.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.5, Part A.
- [ ] Complete the rest of Workbook Lesson 2.5 (Parts B–E).

1 Get Ready

I WONDER

You can see a book in a lit room, but not in a completely dark closet. What is missing in the closet?

2 Learn

A light source gives off its own light. The sun, a lamp, a fire, and a firefly are light sources. Most things, like a book or your hand, do not make light. We see them only because

WORDS & MORE

WORDS TO KNOW

light source
Something that gives off its own light.

reflect
To bounce off a surface.

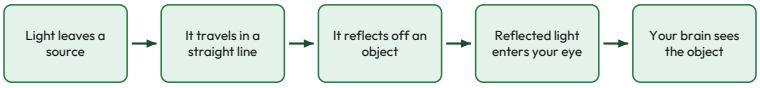
shadow
A dark area made when an object blocks light.

pupil
The opening in the center of the eye that lets light in.

REMEMBER

light reflects, or bounces, off them.

Light travels in straight lines from its source. When light hits an object, some of it bounces off. If that reflected light travels into your eye through your pupil, your eye sends a message to your brain, and you see the object. In a dark closet there is no light to reflect, so you can't see anything.



Because light travels in straight lines, it can't bend around an object that blocks it. This makes a shadow. A shadow is longer when the light comes from low on the side, like the sun in the early morning, and shorter when the light is high overhead, like the sun at noon.

WORDS & MORE

KEY IDEA

Light travels in straight lines. We see an object when light reflects off it and enters our eyes. Blocked light makes shadows.

WATCH OUT

Your eyes don't send out anything to see. Light must travel into your eyes from the object.

✓ REMEMBER

Light travels in straight lines; we see when reflected light enters our eyes.

INVESTIGATE · SHADOW LENGTHS

How does the position of a light change a shadow?

You need: a flashlight; a small toy or a pencil standing in clay; a ruler; a dark room.

1. Stand the toy on a table and shine the flashlight from straight above. Measure the shadow.
2. Shine it from a medium angle. Measure the shadow.
3. Shine it from low on the side. Measure the shadow.
4. Keep the flashlight the same distance away each time.

Record: a table of light position and shadow length, and a model drawing with straight-line arrows.

REMEMBER

WORDS & MORE

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 2.5, Part A. Then check with your teacher.

1. Name two light sources.
2. Why can't you see a book in a dark closet?
3. When is your shadow longest: morning or noon?

REMEMBER

LESSON 2.6
Week 3 · Lesson 2 of 2
Unit 2: Energy, Heat, and Light

Lesson 2.6: Reflection, Absorption, and Transparency

TODAY'S GOAL

I can describe how light reflects, passes through, or is absorbed by different materials.

Standards: 4-PS4-2

TODAY'S ASSIGNMENT

- [ ] Read Lesson 2.6 in this book with your teacher.
- [ ] Do the Investigate activity: Light Detective.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.6, Part A.
- [ ] Complete the rest of Workbook Lesson 2.6 (Parts B–E).

1 Get Ready

I WONDER

Why can you see through a window but not through a wall?

2 Learn

When light hits a material, three things can happen. It can reflect (bounce off), it can pass through, or it can be absorbed (taken in). Most materials do some of each.

WORDS & MORE

WORDS TO KNOW

transparent
Lets almost all light pass through, so you can see clearly through it.

translucent
Lets some light pass through, but things look blurry.

opaque
Blocks light from passing through.

absorb
To take in, as when a material soaks up light.

REMEMBER

Kind of material	What light does	Examples
transparent	passes through almost completely	clear glass, clean water, plastic wrap
translucent	some passes through, scattered	wax paper, frosted glass, a thin sheet
opaque	is blocked: reflected or absorbed	wood, metal, cardboard, your hand

Smooth, shiny surfaces like mirrors reflect light in a neat way, so you can see an image. Light hits a mirror and bounces off at the same angle it came in, like a ball bouncing off a wall. Rough or dull surfaces scatter light in many directions.

Dark materials absorb more light than light-colored ones. Absorbed light turns into heat. That's why a black T-shirt feels hotter in the sun than a white one.

WORDS & MORE

KEY IDEA

Light can reflect, pass through, or be absorbed. Transparent materials let light through, translucent ones let some through, and opaque ones block it.

INVESTIGATE · LIGHT DETECTIVE

Which materials let light through?

You need: a flashlight; materials: clear plastic, wax paper, foil, cardboard, tissue paper, a mirror, a glass of water.

1. Predict whether each material is transparent, translucent, or opaque.
2. Shine the flashlight through each material onto a wall.
3. Record what you see on the wall: bright light, dim blurry light, or no light.
4. Shine the light at the mirror and find where the reflection lands.

Record: a three-column chart: Material | Prediction | Result.

✓ REMEMBER

Light reflects, passes through, or is absorbed; materials are transparent, translucent, or opaque.

REMEMBER

WORDS & MORE

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 2.6, Part A. Then check with your teacher.

1. Is wax paper transparent, translucent, or opaque?
2. What happens to light that hits a mirror?
3. Why does a black shirt feel hotter in the sun?

REMEMBER

LESSON 2.7
Week 4 · Lesson 1 of 2
Unit 2: Energy, Heat, and Light

Lesson 2.7: Refraction and Color

TODAY'S GOAL

I can explain that light bends when it passes into a new material and that white light is made of many colors.

Standards: 4-PS4-2

TODAY'S ASSIGNMENT

- [ ] Read Lesson 2.7 in this book with your teacher.
- [ ] Do the Investigate activity: Make a Spectrum.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.7, Part A.
- [ ] Complete the rest of Workbook Lesson 2.7 (Parts B–D).

1 Get Ready

I WONDER

Put a pencil in a glass of water and look from the side. Why does it look broken?

2 Learn

Light travels at different speeds in different materials. When it passes from air into water or glass at an angle, it changes speed and bends. This bending is called refraction. It makes a pencil in water look broken and makes a pool look shallower than it really is.

WORDS & MORE

WORDS TO KNOW

refraction
The bending of light as it passes from one material into another.

prism
A clear, shaped object that splits white light into colors.

spectrum
The band of colors in white light: red, orange, yellow, green, blue, and violet.

lens
A curved, clear object that bends light.

REMEMBER

A lens is a curved piece of clear glass or plastic that bends light on purpose. Eyeglasses, magnifying glasses, cameras, and the lens inside your own eye all use refraction to focus light.

Sunlight looks white, but it is really a mix of colors. A prism refracts each color a slightly different amount, spreading white light into a spectrum: red, orange, yellow, green, blue, and violet. A rainbow forms when raindrops act like tiny prisms.

Why does a red apple look red? The apple absorbs most colors of light and reflects mostly red light into your eyes. A white object reflects all colors; a black object absorbs almost all of them.

WORDS & MORE

KEY IDEA

Refraction is the bending of light when it passes into a new material. Prisms and raindrops separate white light into a spectrum. We see an object's color from the light it reflects.

INVESTIGATE · MAKE A SPECTRUM

Can you split white light into colors?

You need: a clear glass of water; a sheet of white paper; a sunny window or a bright flashlight.

1. Set the glass of water on a sunny windowsill.
2. Hold the white paper on the floor or table where light passes through the water.
3. Move the paper until you see colors.
4. Record the order of the colors you see.

Record: a colored drawing of your spectrum with the colors in order.

✓ REMEMBER

Light bends when it changes materials; white light is a mix of colors.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 2.7, Part A. Then check with your teacher.

1. Why does a pencil in water look broken?
2. What colors are in white light?
3. Why does a banana look yellow?

REMEMBER

LESSON 2.8
Week 4 · Lesson 2 of 2
Unit 2: Energy, Heat, and Light

Lesson 2.8: Light Investigation: Claim, Evidence, and Reasoning

TODAY'S GOAL

I can plan an investigation about light and explain my results with a claim, evidence, and reasoning.

Standards: 4-PS4-2, Science and Engineering Practices 3, 7

TODAY'S ASSIGNMENT

- [ ] Read Lesson 2.8 in this book with your teacher.
- [ ] Do the Investigate activity: Build a Periscope.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.8, Part A.
- [ ] Complete the rest of Workbook Lesson 2.8 (Parts B–E).

1 Get Ready

I WONDER

Submarine sailors look above the water without coming up. How might they do it?

2 Learn

Scientists answer questions by planning investigations, collecting evidence, and explaining their results. A strong explanation has three parts: a claim, evidence, and

WORDS & MORE

WORDS TO KNOW

periscope
A tube with mirrors that lets you see over or around things.

claim
A statement that answers a question.

evidence
Data and observations that support a claim.

reasoning
The science idea that explains why the evidence supports the claim.

REMEMBER

reasoning (CER).

Part	What it does	Example
claim	answers the question	A periscope lets you see over a wall.
evidence	data you collected	I saw the clock across the room while I was hiding behind the couch.
reasoning	the science idea that connects them	Light travels in straight lines and reflects off mirrors at an angle, so the mirrors carried the clock's light down the tube to my eye.

A periscope uses two mirrors set at an angle. Light from an object enters the top opening, reflects off the top mirror straight down the tube, reflects off the bottom mirror, and travels out into your eye.

KEY IDEA

Scientists explain results with a claim, evidence, and reasoning. A periscope works because light travels in straight lines and reflects off mirrors.

INVESTIGATE · BUILD A PERISCOPE

Can mirrors let you see over a wall?

You need: a long cardboard box (like a juice or milk carton, cleaned); two small mirrors; tape; scissors (with an adult).

1. Cut a square window near the top of one side and near the bottom of the opposite side.
2. Tape one mirror inside the top, tilted to face down the tube; tape the other inside the bottom, tilted to face the bottom window.
3. Look through the bottom window and adjust the mirrors until you see out the top.
4. Hide below a table or couch and test what you can see.

Record: a labeled diagram with arrows showing the light path, and a CER paragraph.

Safety: An adult should help with cutting. Handle mirrors carefully.

✓ REMEMBER

Claim, evidence, and reasoning explain results; a periscope uses reflection.

WORDS & MORE

REMEMBER

WORDS & MORE

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 2.8, Part A. Then check with your teacher.

1. What are the three parts of a scientific explanation?
2. How many mirrors does a periscope use?
3. What science idea explains how a periscope works?

UNIT 2 REVIEW

Unit 2 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
2.1	Energy makes things move or change; faster objects have more motion energy.
2.2	Temperature tells how hot something is; heat flows from warmer to cooler.
2.3	Heat moves by conduction, convection, and radiation.
2.4	Conductors move heat easily; insulators slow it; engineers design with both.
2.5	Light travels in straight lines; we see when reflected light enters our eyes.
2.6	Light reflects, passes through, or is absorbed; materials are transparent, translucent, or opaque.
2.7	Light bends when it changes materials; white light is a mix of colors.
2.8	Claim, evidence, and reasoning explain results; a periscope uses reflection.

Glossary

absorb	To take in, as when a material soaks up light.
air sacs	Tiny balloon-like sacs in the lungs where gases pass into and out of the blood.
artery	A blood vessel that carries blood away from the heart.
capillary	A tiny blood vessel where materials pass between blood and cells.
carbon dioxide	A waste gas your cells make and you breathe out.
cell	The smallest living unit.
circulatory system	The heart and blood vessels that move blood.
claim	A statement that answers a question.
conduction	Heat moving through touching objects.
conductor	Lets heat move easily.
convection	Heat moving by the flow of a liquid or gas, such as air or water.
criteria	What a design must do to succeed.
degree	A unit for measuring temperature, shown with the symbol °.
design	A plan for building something that solves a problem.
diaphragm	A dome-shaped muscle below the lungs that helps you breathe.
digestive system	Organs that break down food.
energy	The ability to make something move or change.
energy source	Something that gives off energy, such as the sun or a battery.
esophagus	The tube that carries food from the mouth to the stomach.
evidence	Observations and data that support a claim.
fair test	Only one variable is changed.
heart	A muscular organ that pumps blood.
heat	Energy flowing from warmer to cooler.
inference	A conclusion based on observations.
insulator	Slows heat.
investigate	To find out about something by observing, testing, and measuring.
joint	A place where two bones meet.
large intestine	The wider tube where water is absorbed from leftover food.

lens	A curved, clear object that bends light.
ligament	A tough band that connects bone to bone.
light source	Something that gives off its own light.
motion energy	The energy an object has because it is moving.
muscular system	Muscles that move the body.
nervous system	The brain, spinal cord, and nerves that control the body.
nutrients	Substances in food the body uses.
observation	Information gathered with senses or tools.
opaque	Blocks light from passing through.
organ	A body part made of tissues that does a special job.
organ system	Organs that work together.
oxygen	A gas in air that cells need to get energy from food.
periscope	A tube with mirrors that lets you see over or around things.
prism	A clear, shaped object that splits white light into colors.
pulse	The beat you can feel as the heart pushes blood through an artery.
pupil	The opening in the center of the eye that lets light in.
radiation	Heat moving through space as waves, such as heat from the sun or a fire.
reasoning	The science idea that explains why the evidence supports the claim.
reflect	Bounce off.
refraction	The bending of light.
respiratory system	Organs that bring in oxygen and remove carbon dioxide.
shadow	A dark area made when an object blocks light.
skeletal system	Bones that support and protect the body.
small intestine	The long, coiled tube where most nutrients pass into the blood.
spectrum	The band of colors in white light: red, orange, yellow, green, blue, and violet.
temperature	How hot or cold something is.
tendon	A tough cord that connects a muscle to a bone.
thermometer	A tool that measures temperature.
tissue	A group of similar cells that work together.
transfer	To move from one place or object to another.

translucent	Lets some light pass through, but things look blurry.
transparent	Lets almost all light pass through, so you can see clearly through it.
variable	Something that can change in an investigation.
vein	A blood vessel that carries blood back to the heart.

Grade 4 Science

Student Textbook

A full year of Grade 4 science built around questions, investigations, and evidence. Students observe, measure, model, and explain the natural world, then use what they learn to design solutions to real problems.

This Student Textbook teaches every lesson: the big idea, key vocabulary, models and examples, and a short set of practice problems to check understanding. It pairs with the Student Workbook and the Teacher Edition.

INSIDE THIS YEAR

01 The Human Body

02 Energy, Heat, and Light

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- Key vocabulary with student-friendly definitions
- Models, examples, and Key Idea summaries
- A study guide and glossary for every unit

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